



COMMERCIAL BANKING, CORP

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REQUEST FOR PROPOSAL

RFP #: PO – S2.H1

TITLE: VOLATILITY MODELING

CLOSING DATE AND TIME: APRIL 2, 2021 @ 5:00 PM

Volatility Modeling: PO – S2.H1

Background and Purpose

By responding to this Request for Proposal (RFP), the Proposer agrees that s/he has read and understood all documents within this RFP package.

Submission Details

Responders to this RFP should supply:

- A business report up to 5 pages, including any supporting plots and tables.

The report should address **all points described in the “Objective” section** below.

The report should be returned in the following way:

- Electronic (submit on Moodle)

Objective

The Commercial Banking Corporation (hereafter the “Bank”), acting by and through its department of *Portfolio Pricing* is seeking proposals for banking services. The scope of services includes the following:

- Collecting of **daily** stock prices and **daily** returns for the 30 stocks in the Dow Jones Industrial Average (DJIA).
- Ranking these 30 stock components of the DJIA by most significant to be predicted with ARCH/GARCH modeling.
 - The Bank’s analysts recommend using the Lagrange Multiplier Lag 1 test p-values to make this ranking; the Bank is open to other proposals as long as they are clearly stated and supported.
- Creation of a series of five models to predict the volatility from the top five most significant stock components of the DJIA to be predicted with ARCH/GARCH modeling.
 - The Bank’s analysts recommend using the top 5 stock components ranked by the LM test previously calculated.
 - The Bank’s analysts recommend picking one of the following models: GARCH(1,1)-Normal, t-GARCH(1,1), QGARCH(1,1)-Normal, QGARCH(1,1)-t; the Bank is open to other proposals as long as they are clearly stated and supported.
 - The Bank’s analysts recommend picking the best model based off of lowest BIC value; the Bank is open to other proposals as long as they are clearly stated and supported.
- Forecast of the next 30 days of volatility for each of the five models mentioned above.
- Rank these five stocks by order of most likely to be bothered by a shock to the market today.
- Rank these five stocks by order of longest effect a shock lasts; compare this to the previous ranking and interpret.

Data Provided

No data is provided to the consulting team for this analysis. All data must be gathered by the team itself through any means it desires. The dates for the data you should pull are March 1, 2011 to March 24, 2021.